Area and perimeter

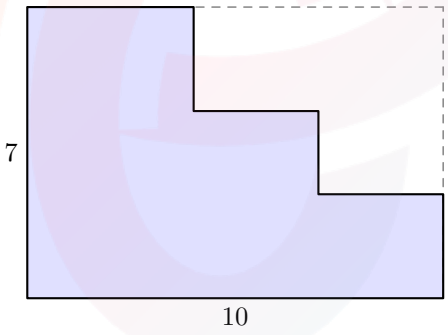
L1-GE2 Level 1 Foundations · Geometry

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How to find perimeters and areas of polygons and circles by formulas, by cutting and subtracting, and with a first use of Pythagoras.
By the end you can find perimeters of shapes with missing lengths, and of shapes that are joined or cut; use the area formulas for triangles, parallelograms, trapezia and circles, and find shaded areas by cutting or subtracting; use Pythagoras' theorem to find a missing length, and recognise the triangles 3, 4, 5 and 5, 12, 13.
Prerequisites: Angle facts and polygons (L1-GE1). **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 120 minutes.

1 Lesson

The shaded shape below has only right angles. It is 10 wide along the bottom and 7 high on the left, and no other lengths are given. Find its perimeter.



We cannot find the length of each step, but we do not need to. The tops of the steps together cover the width of the shape once, so they add to 10, like the bottom edge. Similarly the vertical edges on the right add to 7, like the left edge. So the perimeter is $2 \times 10 + 2 \times 7 = 34$, the perimeter of the dashed rectangle. Pushing the steps out to the corner changes the area of the shape, but not its perimeter.

This unit covers perimeters and areas of polygons and circles, the methods of cutting and subtracting that contest questions are built on, and a first look at Pythagoras' theorem.

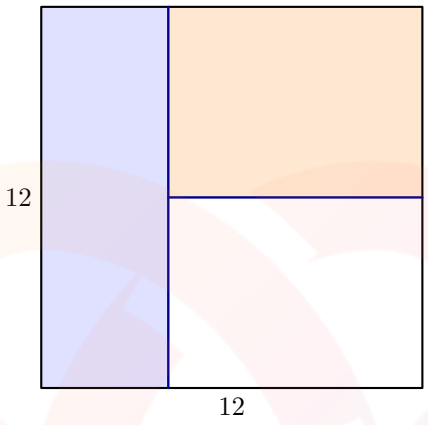
1.1 Perimeter

The perimeter is the length of the boundary. A rectangle with length l and width w has perimeter $2(l + w)$ and area lw.

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Fact 1 (perimeter).
(a) If a shape has only right angles, and every horizontal and every vertical line meets it in a single piece, then its perimeter equals the perimeter of the smallest rectangle around it. (b) If two shapes are joined along an edge of length e, the perimeter of the combined shape is the sum of their perimeters minus $2e$.

Part (a) is the argument of the opening problem. The condition matters: a U shape fails it, and the inside edges of the U add extra length. For (b), the shared edge was counted once in each shape and is no longer on the boundary. Read backwards, (b) says that cutting a shape along a line of length e adds $2e$ to the total perimeter of the pieces.



Example 1.
A square of side 12 is cut into three rectangles of equal area, as shown: one cut goes from top to bottom, and a second cut divides the larger piece into two. Find the sum of the perimeters of the three rectangles.

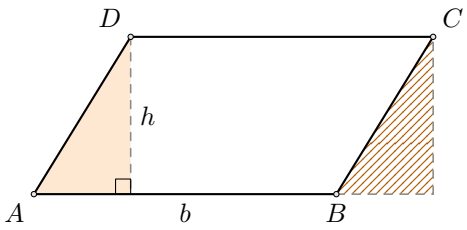
Solution.
Each rectangle has area $\frac{1}{3} \times 144 = 48$. The first rectangle is 12 tall, so it is $48 \div 12 = 4$ wide. The rest of the square is 8 by 12, and it is cut into two equal rectangles, each 8 by 6. The perimeters are 32, 28 and 28, a total of 88.
Fact 1(b) gives this without the separate perimeters. The outside of the square is counted once and each cut is counted twice, once for each piece beside it, so the total is $48 + 2 \times (12 + 8) = 88$. Use this method when the pieces have awkward sizes.

1.2 Triangles, parallelograms and trapezia

Fact 2 (area formulas).
A parallelogram with base b and height h has area bh. A triangle with base b and height h has area $\frac{1}{2}bh$. A trapezium whose parallel sides a and b are a distance h apart has area $\frac{1}{2}(a+b)h$.

The height is measured at right angles to the base, never along a slanted side. Cutting the right-angled triangle off one end of a parallelogram and moving it to the other end gives a b by h rectangle, as in the diagram.

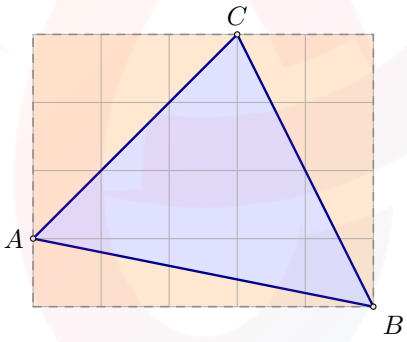
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Two copies of a triangle, the second turned half round, make a parallelogram with the same base and height, so the triangle has area $\frac{1}{2}bh$. In the same way, two copies of a trapezium make a parallelogram with base $a + b$ and height h. Any side of a triangle can be the base, with the height measured from the opposite vertex; in an obtuse triangle this height may fall outside the triangle. It follows that triangles with the same base and the same height have equal areas: sliding the top vertex along a line parallel to the base does not change the area.

1.3 Cutting, rearranging and subtracting

For a region that is not a standard shape, either cut it into pieces whose areas you know and add them, or take a larger simple shape and subtract the parts that are not in the region. Decide which before calculating anything.



Example 2.
Triangle ABC is drawn on a grid of unit squares, as shown. Find its area.

Solution.
No side of the triangle lies along a grid line, so a base and its height are hard to find. Instead, we enclose the triangle in the 5 by 4 rectangle shown and subtract the three corner triangles, whose shorter sides do lie along grid lines.
The corner triangles have shorter sides 5 and 1, 3 and 3, and 2 and 4, so their areas are $\frac{5}{2}$, $\frac{9}{2}$ and 4. The rectangle has area 20, so triangle ABC has area $20 - \frac{5}{2} - \frac{9}{2} - 4 = 9$.
This works for any triangle on a grid: draw the smallest rectangle around it and remove the corner pieces, which are right-angled triangles (sometimes together with rectangles).

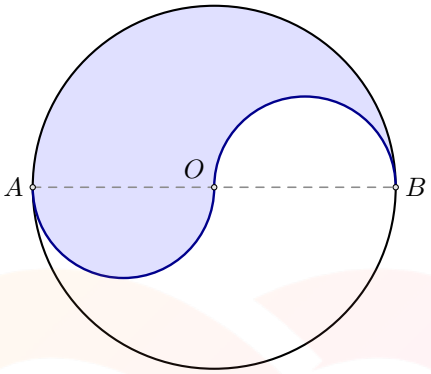
1.4 Circles

For every circle, the circumference divided by the diameter is the same number, called π (about 3.14). So a circle with radius r and diameter $d = 2r$ has

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$$\text{circumference} = \pi d = 2\pi r, \qquad \text{area} = \pi r^2.$$

For the area, cut the circle into thin slices through the centre and lay them in a row, pointing alternately up and down. They form nearly a rectangle of height r and length πr (half the circumference), and the thinner the slices, the closer it is; so the area is $\pi r \times r = \pi r^2$. Answers are usually left in terms of π. A semicircle of radius r has area $\frac{1}{2}\pi r^2$ and a curved edge of length πr; its perimeter also includes the straight edge.



Example 3.
A circle has centre O and diameter $AB = 12$. A semicircle on AO is drawn below AB, and a semicircle on OB is drawn above AB. Together they form an S-shaped curve from A to B, which splits the circle into two pieces. Find the area and the perimeter of the shaded piece.

Solution.
The piece is built from standard shapes: the upper half of the big circle, which has area $\frac{1}{2} \times \pi \times 6^2 = 18\pi$, minus the semicircle on OB, plus the semicircle on AO. Both small semicircles have radius 3, so what is removed equals what is added, and the area of the piece is 18π. (Alternatively, a half turn about O swaps the two pieces, so each is half the circle.)
The boundary of the piece is half the big circle, of length $\frac{1}{2} \times 2\pi \times 6 = 6\pi$, and the two small semicircles, each of length 3π. So the perimeter is 12π. That is the whole circumference of the big circle, although the area is only half of it.

1.5 A first look at Pythagoras' theorem

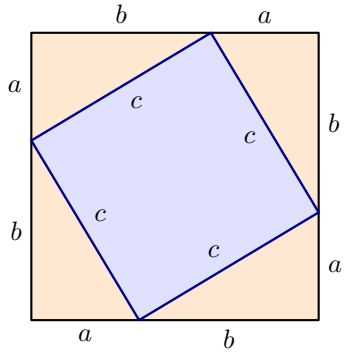
Fact 3 (Pythagoras' theorem).
In a right-angled triangle with shorter sides a and b and longest side c (the *hypotenuse*, opposite the right angle), $a^2 + b^2 = c^2$.

Proof. Place four copies of the triangle in the corners of a square of side $a + b$, as shown. Where two triangles meet on a side of the big square, their angles are the two acute angles of the triangle, which add to $90°$ (the third angle of the triangle is the right angle), so the middle shape has an angle of $180° - 90° = 90°$ there. It has four sides of length c and four right angles, so it is a square of area c^2, and

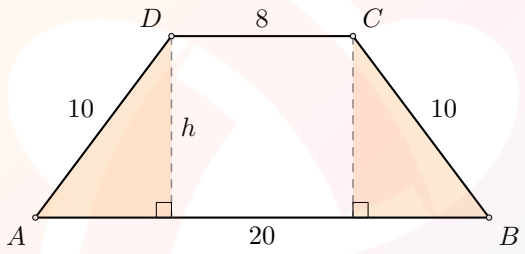
$$(a+b)^2 = c^2 + 4 \times \tfrac{1}{2}ab = c^2 + 2ab.$$

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Cutting the big square instead into an a by a square, a b by b square and two a by b rectangles gives $(a+b)^2 = a^2 + b^2 + 2ab$. Comparing the two, $a^2 + b^2 = c^2$.



Recognise the whole-number triangles $3, 4, 5$ and $5, 12, 13$, and multiples such as $6, 8, 10$; setters choose them on purpose.



Example 4.
$ABCD$ is a trapezium with AB parallel to DC, $AB = 20$, $DC = 8$ and $AD = BC = 10$. Find its area.

Solution.
The formula needs the height h, which is not given. Heights come with right angles, so drop perpendiculars from D and C to AB. The middle part of AB is opposite DC in a rectangle, so it is 8, and the two end parts add to 12. The two end triangles are right-angled with the same hypotenuse 10 and the same height h, so by Pythagoras their bases are equal, and each is 6.
Then $6^2 + h^2 = 10^2$, so $h^2 = 64$ and $h = 8$ (the triangle is twice $3, 4, 5$). The area is $\frac{1}{2}(20 + 8) \times 8 = 112$. Check by cutting: the rectangle has area $8 \times 8 = 64$ and each end triangle has area $\frac{1}{2} \times 6 \times 8 = 24$, and $64 + 24 + 24 = 112$.

Common traps.

- Using a slanted side as the height.
- Confusing radius with diameter, or circumference $2\pi r$ with area πr^2.
- Leaving out the straight edges of a semicircle or quarter circle in a perimeter.
- Assuming that shapes with equal areas have equal perimeters, or the reverse.
- Using Pythagoras in a triangle with no right angle, or with the wrong side as the

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hypotenuse.

1.6 How to spot this in a contest

1. If a perimeter is asked for and lengths are missing, push edges outwards, or use Fact 1(b) for shapes that are joined or cut.
2. For an awkward region, decide whether to add pieces or to subtract from a rectangle or circle around it.
3. If a shape on a grid has sides off the grid lines, enclose it in a rectangle and subtract.
4. For curved regions, look for equal pieces that can be moved to fill gaps.
5. If a length is missing, find or make a right angle and use Pythagoras, trying $3, 4, 5$ and $5, 12, 13$ first.
6. If no answer choice contains π, the curved parts of the region must cancel.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ A 2 by 3 rectangle is cut from one corner of a 9 by 5 rectangle. Find the area and the perimeter of the shape that is left.

Problem 2 $\star$ $ABCD$ is a rectangle with $AB = 10$ and $BC = 6$, and P is a point on the side CD. Find the total area of triangles APD and BPC.

Problem 3 $\star\star$ A semicircle is drawn outwards on each side of a square of side 6, making a shape with four rounded parts. Find the perimeter and the area of this shape, in terms of π.

Problem 4 $\star\star$ Square $ABCD$ has side 17. Points P, Q, R and S lie on the sides AB, BC, CD and DA respectively, with $AP = BQ = CR = DS = 5$. Find the area of $PQRS$.

Problem 5 $\star\star\star$ $ABCD$ is a trapezium with AB parallel to DC, and its diagonals AC and BD meet at X. Explain why triangles AXD and BXC have the same area.

Problem 6 $\star\star\star$ A circle has radius 5. A square is drawn inside it with all four of its corners on the circle. Find the total area of the four regions that are inside the circle but outside the square, in terms of π.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 8 2016 Problem 2** $\star$
   Choose a base along a side of the rectangle.

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2. **UKMT Junior Kangaroo 2017 Question 7** ⋆
   What does the cut add to the total perimeter?
3. **AMC 8 2019 Problem 2** ⋆
   Compare the long and short sides of the small rectangles.
4. **UKMT Junior Kangaroo 2019 Question 5** ⋆
   The black region belongs to both rectangles.
5. **UKMT Junior Kangaroo 2018 Question 13** ⋆⋆
   What happens to the edge that the two shapes share?
6. **UKMT JMC 2017 Question 8** ⋆⋆
   Compare the triangle with the rectangle on the same base.
7. **UKMT Junior Kangaroo 2019 Question 6** ⋆⋆
   Which diagonal splits the quadrilateral into two right-angled triangles?
8. **AMC 8 2018 Problem 15** ⋆⋆
   Doubling the radius multiplies the area by four.
9. **UKMT Junior Kangaroo 2017 Question 15** ⋆⋆
   The unshaded triangles all have the same height.
10. **AMC 8 2017 Problem 18** ⋆⋆⋆
    Draw BD and look for familiar right-angled triangles.
11. **UKMT Junior Kangaroo 2019 Question 19** ⋆⋆⋆
    Find the size of the small squares first.
12. **UKMT JMC 2019 Question 23** ⋆⋆⋆
    Extend lines to make a rectangle, then subtract.

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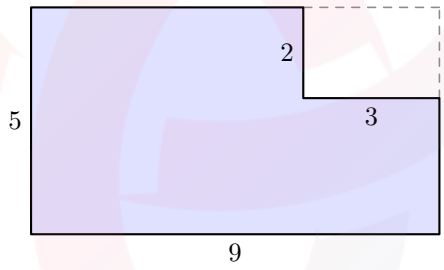
4 Hints

1. For the perimeter, compare the two new edges with the parts of the old edges that were removed.
2. First find the area of triangle ABP. What is its height?
3. Each semicircle has radius 3. The sides of the square are inside the shape, so they are not part of its perimeter.
4. Subtract the four corner triangles from the big square.
5. Compare triangles ABD and ABC. They have the same base. What about their heights?
6. Draw the diagonals of the square. Where do they cross, and how long are their halves?

5 Solutions

Problem 1

A 2 *by* 3 *rectangle is cut from one corner of a* 9 *by* 5 *rectangle. Find the area and the perimeter of the shape that is left.*



The area is $9 \times 5 - 2 \times 3 = 45 - 6 = 39$.

For the perimeter, the cut removes a piece of length 3 from one side and a piece of length 2 from the next side, but it creates two new edges of lengths 3 and 2 (the other two sides of the small rectangle). So the perimeter does not change, and it is $2 \times (9 + 5) = 28$. This is Fact 1(a): pushing the two new edges out to the corner gives back the 9 by 5 rectangle.

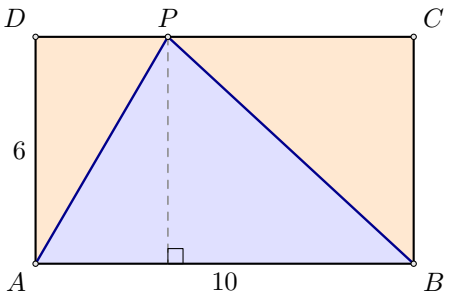
So cutting off the corner reduced the area but not the perimeter. Cutting a notch out of the middle of a side would instead increase the perimeter.

Answer: area 39, perimeter 28

Problem 2

ABCD is a rectangle with $AB = 10$ *and* $BC = 6$*, and P is a point on the side CD. Find the total area of triangles APD and BPC.*

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Take AB as the base of triangle ABP. Its height is the distance from P to AB. Since P lies on CD, which is parallel to AB, this distance is $BC = 6$ wherever P is. So triangle ABP has area $\frac{1}{2} \times 10 \times 6 = 30$, half the area of the rectangle.

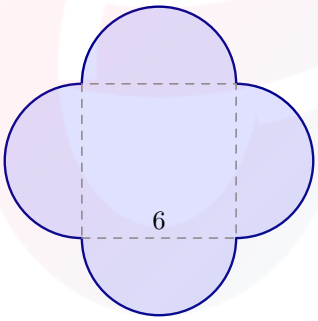
The three triangles APD, ABP and BPC fill the rectangle, which has area 60. Hence triangles APD and BPC have total area $60 - 30 = 30$.

(Directly: if $DP = t$ then $PC = 10 - t$, and the two triangles have areas $\frac{1}{2} \times t \times 6 = 3t$ and $\frac{1}{2} \times (10 - t) \times 6 = 30 - 3t$, which add to 30.)

Answer: 30

Problem 3

A semicircle is drawn outwards on each side of a square of side 6, making a shape with four rounded parts. Find the perimeter and the area of this shape, in terms of π.



Each semicircle has diameter 6 and radius 3. The boundary of the shape consists of the four curved edges only, since the sides of the square are now inside the shape. A whole circle of radius 3 has circumference $2\pi \times 3 = 6\pi$, so each curved edge has length 3π, and the perimeter is $4 \times 3\pi = 12\pi$.

The area is the square plus four semicircles. The square has area 36, and each semicircle has area $\frac{1}{2} \times \pi \times 3^2 = \frac{9}{2}\pi$, so the four of them have area 18π. The total area is $36 + 18\pi$.

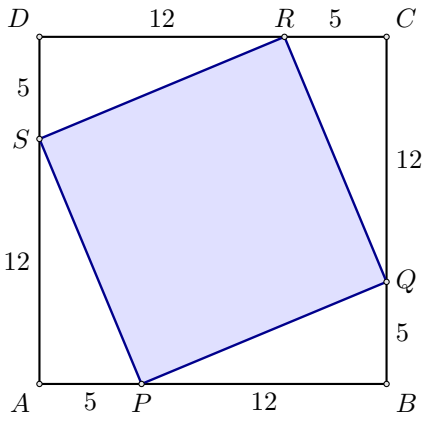
The four semicircles make two whole circles of radius 3, which gives a quick check of both answers: two such circles have total circumference 12π and total area 18π.

Answer: perimeter 12π, area $36 + 18\pi$

Problem 4

Square $ABCD$ has side 17. Points P, Q, R and S lie on the sides AB, BC, CD and DA respectively, with $AP = BQ = CR = DS = 5$. Find the area of $PQRS$.

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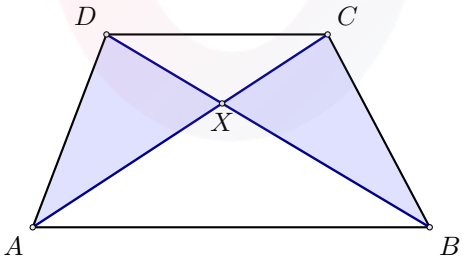
Each corner of the big square contains a right-angled triangle. For example, triangle PBQ has $PB = 17 - 5 = 12$ and $BQ = 5$, with the right angle at B. All four corner triangles have shorter sides 5 and 12, so each has area $\frac{1}{2} \times 5 \times 12 = 30$. The area of $PQRS$ is $17^2 - 4 \times 30 = 289 - 120 = 169$.

In fact $PQRS$ is a square. Its four sides are the hypotenuses of the four corner triangles, so they are equal, and at each of its vertices the two acute angles of the triangles add to $90°$, leaving a right angle. Its side is 13, since $13^2 = 169$, and Pythagoras in triangle PBQ agrees: $5^2 + 12^2 = 25 + 144 = 169$. This is the picture from the proof of Pythagoras' theorem in the lesson.

Answer: 169

Problem 5

$ABCD$ is a trapezium with AB parallel to DC, and its diagonals AC and BD meet at X. Explain why triangles AXD and BXC have the same area.



Consider triangles ABD and ABC. They have the same base AB. The height of triangle ABD is the distance from D to the line AB, and the height of triangle ABC is the distance from C to the line AB. Since DC is parallel to AB, these two distances are equal: both are the distance between the parallel sides. So triangles ABD and ABC have the same area.

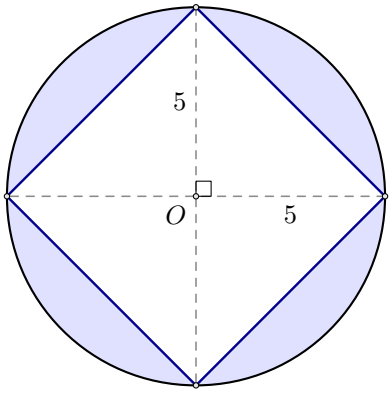
Triangle ABD is made up of triangles AXB and AXD, and triangle ABC is made up of triangles AXB and BXC. Removing the common triangle AXB from both, we find that triangles AXD and BXC have the same area.

The two triangles are usually not congruent (in the diagram they have different shapes), so the equal areas come from the equal bases and heights, not from symmetry.

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Problem 6

A circle has radius 5*. A square is drawn inside it with all four of its corners on the circle. Find the total area of the four regions that are inside the circle but outside the square, in terms of* π*.*



The circle has area $\pi \times 5^2 = 25\pi$, so we need the area of the square. Its side is not given, so we use its diagonals. The diagonals of a square are equal, cross at right angles and cut each other in half, so the point where they cross is the same distance from all four corners. That point is the centre O of the circle, and each half of a diagonal is a radius, of length 5.

The diagonals cut the square into four right-angled triangles, each with shorter sides 5 and 5 and area $\frac{1}{2} \times 5 \times 5 = \frac{25}{2}$. So the square has area $4 \times \frac{25}{2} = 50$. Alternatively, Pythagoras in one of these triangles gives the side s of the square: $s^2 = 5^2 + 5^2 = 50$, and s^2 is the area of the square, so we never need s itself.

The four regions have total area $25\pi - 50$, which is about 28.5.

Answer: $25\pi - 50$